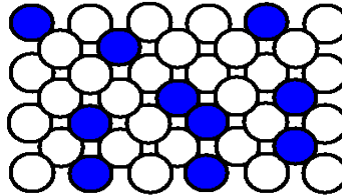




**Politecnico  
di Torino**

Mechanical Engineering  
*Metallic Materials and Technology - MMT*

# ALLOY FORMATION



Metallic Materials and Technology  
Lecture n.3

A.Y. 2025-26  
G. Maizza

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1

1

## Humé-Rothery rules for Alloy formation:

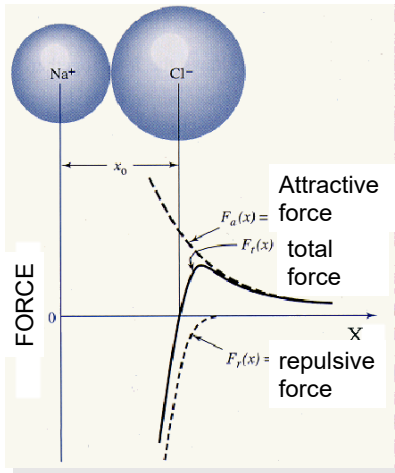
Hume-Rothery (1899-1968), eminent metallurgist working at Oxford University

### The 4 rules of Humé-Rothery for solid solutions

- **Dimensional factor** (atoms tend to pack => the atomic radii of solvent atoms must stay within 15%)
- **Identical crystal structure** of solute and solvent
- **Comparable electronegativity** of both solute and solvent (otherwise the formation of new intermetallic phase takes part)
- **The valence factor**: a solute enters into solution if its valence is greater than that of the solvent

2

## Interatomic Forces: Condon-Morse model



$x_0$  is the equilibrium distance at which attraction and repulsion forces balance to each other ( $\Sigma F=0$ ).

If  $x < x_0$  the **negative repulsion force** due to electrons, prevails;

If  $x > x_0$  the attraction force is dominant.

Figure shows the resultant force (solid line).

The Condon-Morse model can be also presented as a function of the the potential (U)

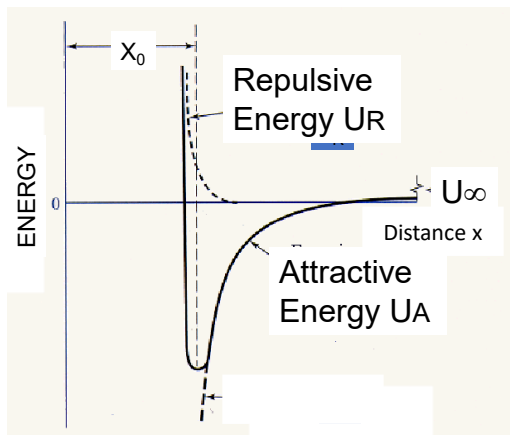
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Lecture 2

3

3

## Interatomic Bonding Energy (potential)



Relationship between Energy and Force:

$$U = \int_{\infty}^r F dx$$

$$= U_A + U_R$$

Equilibrium sets in when the minimum energy is attained.

Van Vlack, p.46

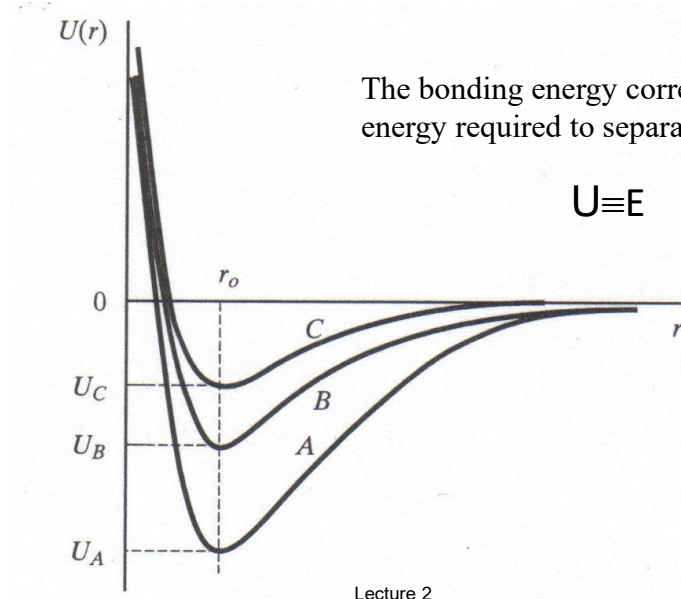
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4

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4

## Bonding energy for three materials



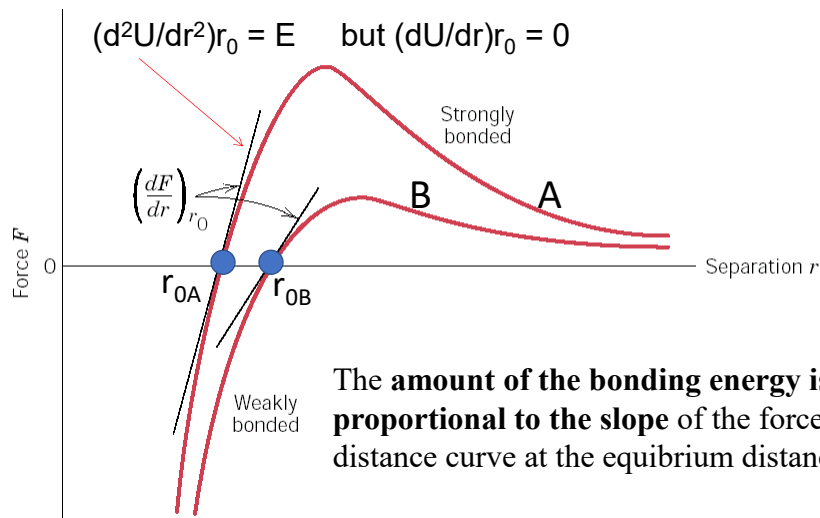
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Lecture 2

5

5

## Elastic modulus E



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Lecture 2

6

6

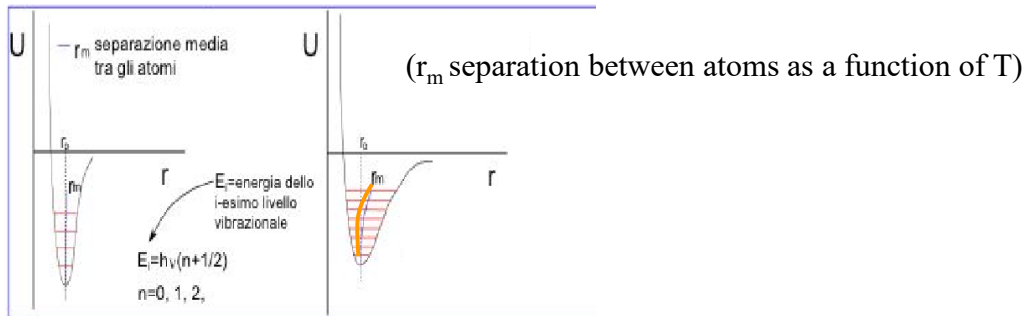
## The elastic modulus: vs. T

$$E = \frac{1}{r_m} \left( \frac{d^2 U}{dr^2} \right)_{r_m} = \text{modulo di Young}$$

The equilibrium separation distance  $r_m$  increases with increasing temperature, thus the potential energy reduces its concavity

$r_0$  : equilibrium distance between atoms at 0K

$r_m$ : average distance between atoms as a function of T



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Lecture 3

7

7

## Elastic Modulus E: mathematical proof

$$F = \frac{dU}{dr}$$

$$U = U(r) = (\text{Sviluppo di Taylor}) = U_0 + \left( \frac{dU}{dr} \right)_{r_0} dr + \frac{1}{2} \left( \frac{d^2 U}{dr^2} \right)_{r_0} dr^2 + \frac{1}{6} \left( \frac{d^3 U}{dr^3} \right)_{r_0} dr^3 + \dots$$

$$\text{per } r = r_0 \rightarrow \frac{dU}{dr} = 0 \text{ because of } U_{\min}$$

$$U(r) = U_0 + \frac{1}{2} \left( \frac{d^2 U}{dr^2} \right)_{r_0} dr^2$$

$$F = \frac{dU}{dr} = \left( \frac{d^2 U}{dr^2} \right)_{r_0} dr$$

$$\sigma = \frac{F}{A} = \frac{F}{\pi r_0^2}$$

$$\varepsilon = \frac{dr}{r_0}$$

$$\sigma = \left( \frac{d^2 U}{dr^2} \right)_{r_0} \frac{dr}{r_0^2} = E \varepsilon \quad E = \frac{1}{r_m} \left( \frac{d^2 U}{dr^2} \right)_{r_m} = \text{Young modulus or elastic modulus}$$

$r_m$  increases with temperature thereby the potential energy has reduced concavity

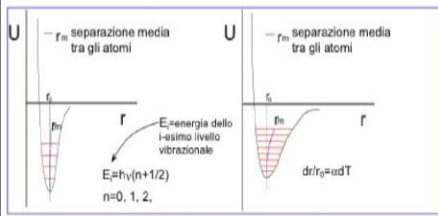
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8

8

## Thermal expansion coefficient



$r_m$  = Average Bonding separation  $f(T)$

$r_0$  = Equilibrium distance at 0K

$r_{m1} > r_{m2}$  expansion  
 $r_{m1} < r_{m2}$  contraction

$$\alpha_L = \frac{1}{L} \frac{dL}{dT}$$

Linear expansion coefficient

$$\alpha_V = \frac{1}{V} \frac{dV}{dT} = 3\alpha_L$$

Volumetric expansion coefficient  
 (for isotropic materials)

$$\alpha_V = \frac{1}{V} \frac{dV}{dT} = \alpha_x + \alpha_y + \alpha_z$$

Volumetric expansion coefficient  
 (for anisotropic materials)

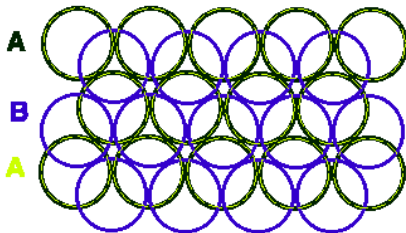
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Lecture 3

10

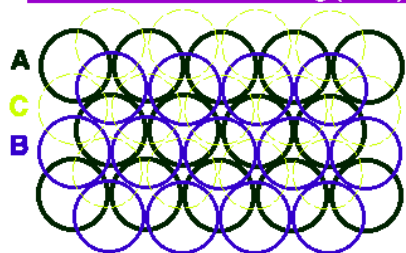
10

## 3 Layers Superposition of 3 planes: closest packing (fcc)



ABA Hexagonal Close-Packing (HCP)

ABC Cubic Close-Packing (CCP)



### TO BE NOTED

- Do not confuse atoms with lattice sites
- The lattice sites are infinitesimal points in 3D space
- Atoms are physical entities
- Lattice sites may not necessarily coincide with the center of the atoms

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11

11

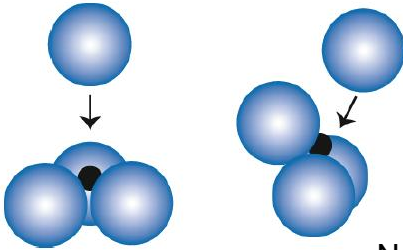
## Interstitial sites

There exist two types of interstitial lattice sites:

Octahedral (coord. N.6)

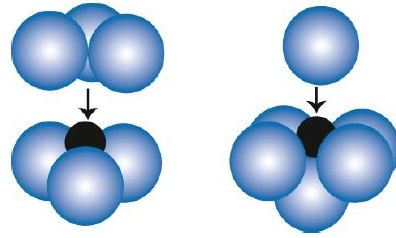
Tetrahedral (coord. N.4)

tetrahedral



Compact structures

octahedral



N. of tetrahedral sites =  $2n$

N. of octahedral sites =  $n$

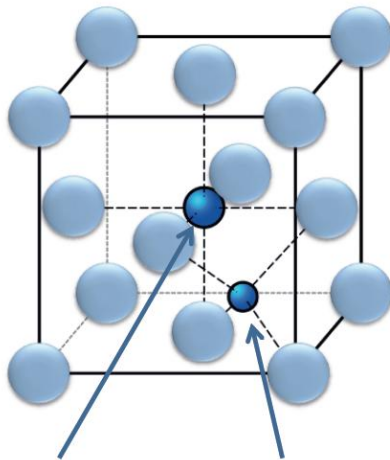
$n$  = n. of atoms in the unit cell

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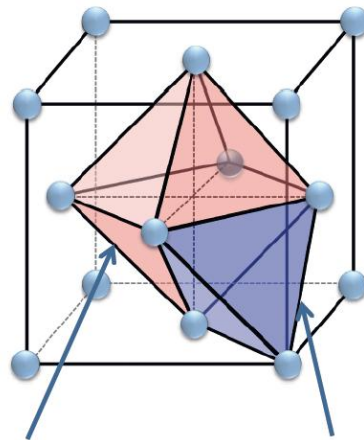
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12

## Interstitials sites in FCC crystals



Octahedral Tetrahedral



Octahedral Tetrahedral

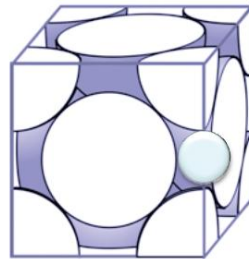
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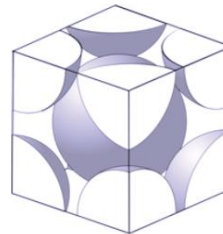
13

## Interstitial sites in Fe lattice and C

Two different packing factors (P.F.): (BCC) = 0.68 and (FCC) = 0.74



**Fe-γ  
(fcc)**



**Fe-α  
(bcc)**

$V_{\text{interst}}(\text{FCC}) > V_{\text{interst}}(\text{BCC})$ , but  $N_{\text{interst}}(\text{FCC}) < N_{\text{interst}}(\text{BCC})$

$T_{\text{amb}} \Rightarrow \text{Fe}_\alpha(\text{bcc}) \text{ 0.02-0.05\%C} \Rightarrow \text{ferrite}$   
 $T_{\text{elevated}} \Gamma (>910^\circ\text{C}) \Rightarrow \text{Fe}_\gamma(\text{fcc}) \text{ 2\%C} \Rightarrow \text{austenite}$

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Lecture 3

14

## GRAIN BOUNDARIES IN POLYCRYSTALLINE MATERIALS

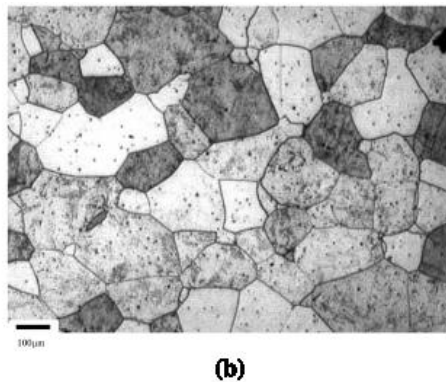
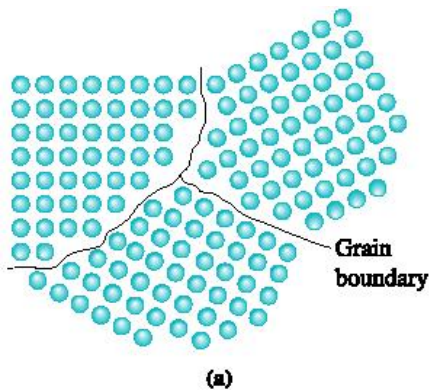


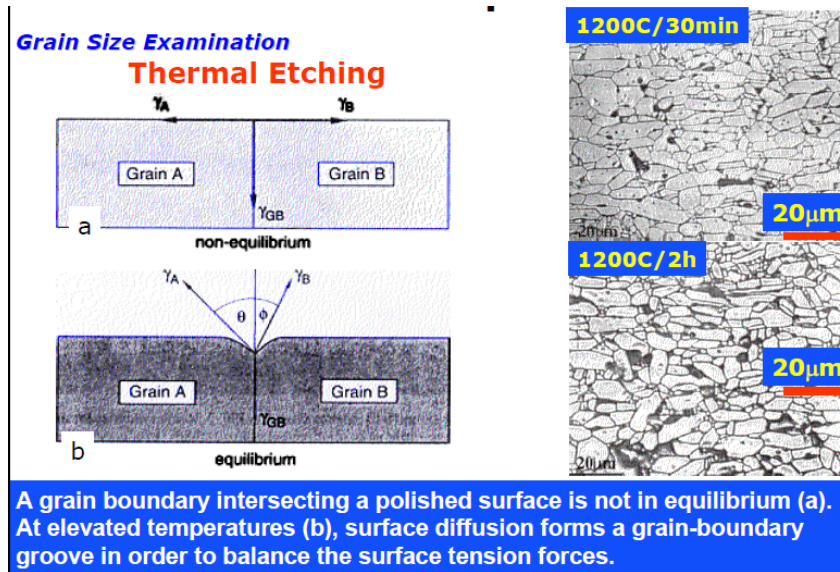
Figure 4.16 (a) The atoms near the boundaries of the three grains do not have an equilibrium spacing or arrangement. (b) Grains and grain boundaries in a stainless steel sample. (Courtesy Dr. A. Deardo.)

15

15

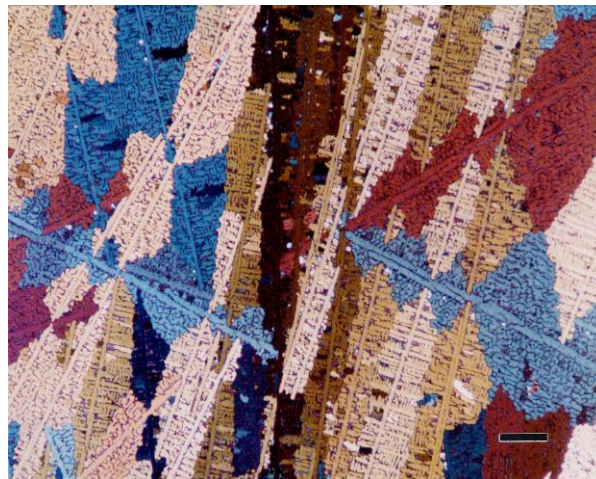


# Thermal Etching and Metallography



16

## Cd-20%Bi cast alloy

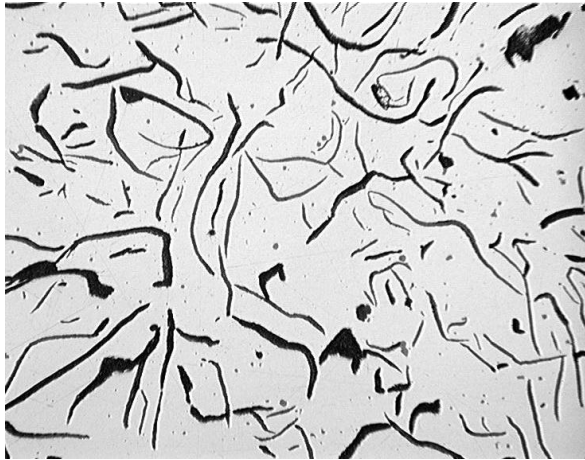


Microstructure not etched, observed with polarized light plus a sensitive dye that reveals the Cd dendrites in various orientations. The interdendritic constituents are the Cd-Bi eutectoid, but it is too fine to resolve at these magnifications.

17



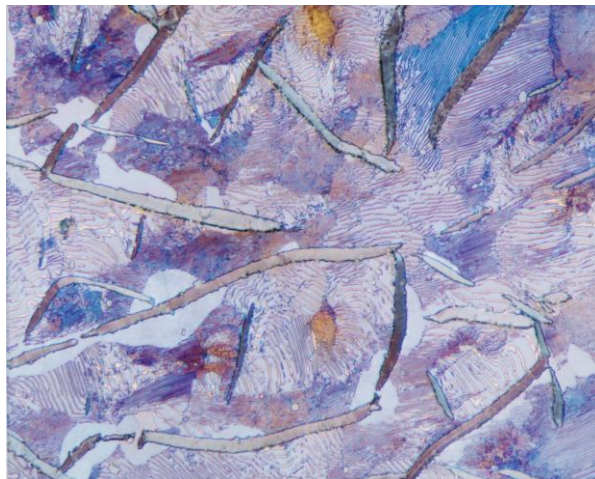
## Second phase morphology in gray cast iron



Graphite is the most stable form of C in ferrous alloys but is only present in cast irons with high C and Si content. In cast irons, graphite can be controlled to produce a variety of shapes from flakes to lamellae to nodules.

18

## Gray cast iron with pearlitic matrix: color etching



Beraha Etchant 500x

19

## Crystal and Microstructure Anisotropy

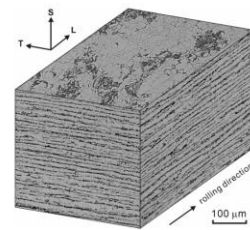
Different directions in a crystal promote different modes of packings

FCC structures show a close packing along the cube diagonal which is different to that along the diagonal of the cube face.

This results in **anisotropic properties** through a single crystal:  
e.g. the strain depends on the direction along which the load is applied.

**In polycrystalline materials, the random orientation of grains tend to make all properties uniform thereby becoming isotropic**

Some polycrystalline materials show oriented grains along working directions (**texture or banding**),  
=> the mechanical properties are affected by the texture orientation



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Lecture 2

20

20

End of Lecture n.3

21

# Properties of metals

- **Intrinsic props:** related to atom bonding nature
- **Extrinsic props:** affected by either the intentional addition of impurities (e.g. Cr, Ni, Co, etc. in steels) or by microstructure as a consequence of materials processing



## Microstructure:

multiscale, defects, anisotropy, ...

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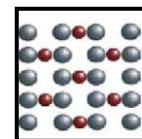
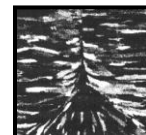
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22

22

## Properties of Metals: classification

- Non-sensitive to microstructure (**intrinsic props.**)
  - density
  - thermal conductivity, (magnetic susceptibility)
  - Elastic modulus
  - thermal expansion coefficient
- Sensitive to microstructure (**extrinsic props.**)
  - Yielding strength
  - Hardness
  - Elongation
  - Impact energy (resilience)
  - Toughness and fracture toughness
  - Fatigue resistance, Creep resistance
  - Corrosion and Oxidation resistance
- Very sensitive to impurities/nanosopic defects
  - Electrical resistivity



19/11/2025

Lecture 3

23

23